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	Title: Primary Immuno Deficiency (PID)	

❖ MEDICAL HISTORY:

* Age ofOnset:

* 4-5 months- Combined T/B cell deficiency

* 6-9 months- B cell deficiency

* Sites ofInfection

* Sinopulmonary infections; otitis media,sinusitis, pneumonia, meningitis, osteomyelitis ..etc -suggest B cell deficiency or combined T & B cell deficiency, neutrophil andcomplement defects.

* Gingivitis, skin infections, Organ abscesses -- suggest Neutrophil/phagocytedefects.

In addition: Delayed separation of theumbilical cord beyond 6 to 8 weeks of age in neonates can suggest a leukocyteadhesion deficiency.

* Microbiologyof the Infections

* Viruses (herpes, varicella, CMV)

o EBV

T cell deficiency

XLP (X linkedlymphoproliferative disorder)

* Fungi

o Candida

Tcell deficiency

- o Pneumocystiscarinii Tcell deficiency

* Parasites

- o Giardialamblia Bcell deficiency
- o Toxoplasma gondii Tcell deficiency

* Bacteria

- o Mycobacteria Tcell deficiency
- o Encapsulated organisms Bcell or complement deficiency

In addition: Patients with XLA have an increased susceptibility to infection with enteroviruses (echovirus, coxsackievirus), which can lead to meningoencephalitis. Arthritis of the large joints can be caused by *Ureaplasma urealyticum*.

* Gastrointestinal Disturbances

- o Malabsorption, diarrhea (G. lamblia) T cell or B cell deficiency
- o chronic viral enteritis (enteroviruses & CMV) B cell or T cell defects.
- o Celiac disease combined B & T cell defects
- o Protracted diarrhea IPEX

* Skin:

- o Refractory eczema Omenn syndrome, Job syndrome & WAS

* Adverse Reactions to Vaccines and Transfusions

- o Disseminated mycobacterial disease after BCG immunization can be seen in interferon-gamma (IFN- γ)-associated & interleukin-12 (IL-12)-related immune deficiencies.

* Family History

- o Consanguinity raises the possibility of an autosomal recessive disorder.
- o Family history of increased autoimmune diseases (such as pernicious anemia, rheumatoid arthritis, systemic lupus erythematosus (SLE), or autoimmune hematologic diseases, lymphoreticular malignancy (leukemia, lymphoma))
- o Neonatal death

❖ PHYSICAL EXAMINATION

Skin and Oral Mucosa

Eczema	Wiskott-Aldrich syndrome, hyper-IgE
Erythrodermic desquamative skin	Omenn syndrome
Petechiae	Wiskott-Aldrich syndrome
Pyoderma, abscesses	Neutrophil or B cell defects
Candidiasis	T cell or combined T/B cell deficiency
Telangiectasia	Ataxia-telangiectasia
Delayed umbilical separation	Neutrophil adhesion defect
Silvery hair, pale skin	Chediak Higashi Syndrome, Griscelli syndrome, HPS type II

Ear, Nose, and Throat

Chronic otitis media: dull tympanic membranes; poor light reflex, scarring, tympanic perforations	B cell deficiency
	Mannose-binding lectin deficiency
Sinusitis: purulent nasal discharge, purulent postpharyngeal exudate, pharyngeal "cobblestoning"	B cell deficiency

Respiratory Tract

Digital clubbing, rales	Defect in any immune component
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Lymphatic System

Absent tonsils/nodes	Bruton's disease, combined T/B cell deficiency
Diffuse lymphoid hyperplasia	CVID, HIV infection

Musculoskeletal System

Arthralgia	B cell deficiency
Dermatomyositis	B cell or complement deficiency
Lupus-like syndrome	Complement or IgA deficiency

Neurologic System

Ataxia	Ataxia-telangiectasia
Enterovirus encephalitis	B cell deficiency (XLA)

Dysmorphism

Micrognathia, short philtrum, ear anomalies	T cell deficiency "DiGeorge anomaly"
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❖ COMMON TYPES OF PRIMARY IMMUNODEFICIENCY DISORDERS (PIDs)

• COMPLEMENT DEFICIENCIES

Complement deficiencies present as

- (1) Severe, recurrent or invasive infection with encapsulated bacteria (due to loss of the opsonic function of C3 fragments),
- (2) Immune complex disease (because complement is important in the solubilization of immune complexes and their removal from the circulation), or
- (3) Severe, recurrent or invasive infection due to *Neisseria*.

Clinical Characteristics of Complement Deficiencies

Deficient Component	Disease Association
Early classical pathway components: C1, C4, C2	Lupus-like illness Vasculitis
Late complement components: C5-C8	Recurrent or disseminated <i>Neisseria</i> infection
C3	Recurrent septicemia Vasculitis Glomerulonephritis
Mannose-binding lectin	Recurrent RTI and OM in early infancy

Laboratory test:

- o total hemolytic complement activity (CH50), which tests for classic pathway activity, and if these are normal, alternative pathway function should be tested with the AH50.
- o Abnormalities in CH50 or AH50 should be pursued by analysis of specific component concentrations in serum.

• PHAGOCYtic/NEUTROPHIL DEFECTS

Clinical Presentation of Phagocytic Dysfunction Disease

- Infections with minimal or no pus
- Infections with formation of abscesses and granuloma
- Recurrent infections with bacteria of low virulence
- Skin infections, furunculosis, organ abscesses, lymphadenitis

- Delayed separation of umbilical cord

Laboratory test:

- CBC - Absolute granulocyte count
- Cell morphology
- Serum total hemolytic complement (CH50, alternative pathway CH50)
- Flow cytometry using dihydrorhodamine dye
- Flow cytometry for leukocyte adhesion molecules (CD11b)

• T CELL IMMUNODEFICIENCY:

Clinical Characteristics of T Cell Deficiencies

- Onset of symptoms in early infancy (4-5 months)
- Recurrent infections with fungal, viral, and mycobacterial pathogens
- Infections caused by opportunistic organisms (e.g., *Pneumocystis carinii*)
- Failure to thrive, often fatal in childhood
- Fatal infections after live-virus vaccines or bacille Calmette-Guérin vaccination
- Graft-versus-host disease from transfusion of blood products
- Increased incidence of malignancy

Laboratory tests:

Absolute lymphocyte count – lymphopenia in 90% of patient with SCID

Chest radiograph for thymus shadow in newborns

Delayed skin hypersensitivity to recall antigens (currently not available)

Quantification of T cell subsets (Flow cytometry)

Lymphocyte proliferative responses to mitogens & antigens (available in SQUH)

Molecular analysis for specific defects

• B CELL IMMUNODEFICIENCY:

Clinical Characteristics of B Cell Deficiencies

Onset of symptoms delayed until age 6-9 months after maternal antibodies wane

Recurrent infections with encapsulated bacteria

Chronic/recurrent sinusitis, otitis media, pneumonia, bronchiectasis

Lymphadenitis and abscess formation

Increased incidence of allergy and autoimmune diseases

Hypoplasia of peripheral lymph nodes and nasopharyngeal lymphoid tissues (XLA)

Lymphoid hyperplasia/hepatosplenomegaly (CVID)

Laboratory Tests

Quantitative serum immunoglobulins

Specific antibodies (hepatitis B, MMR, varicella, tetanus, diphtheria, Anti Hib, pneumococcal)

Isohemagglutinins (IgM antibodies to A and B blood group antigens) - not available

Lymphocyte subset.

❖ MANAGEMENT OF PIDs:

i. General aspect for management of PID

- a) Low threshold to admit patient diagnosed or suspected to have PID.
- b) Take necessary cultures and start broad spectrum antimicrobials pending culture results.
- c) Be aware that patients with certain types of PID such as severe T cell disorders or innate immune defects do not mount a normal inflammatory response i.e: WBC or CRP might be within normal limits at the time of severe infection.
- d) If a patient with severe T cell disorders encounters an affected patient with chicken pox, then the PID patient should receive varicella zoster immunoglobulin within 10 days of exposure. (please refer to IM VZIG protocol)

ii. Specific management points related to the type of PID

a) Complement defect

- Symptomatic care
- Prophylactic antibiotics
- Immunizations with bacterial polysaccharide vaccines (e.g. Pneumococcal/ Meningococcal vaccines) + annual flu vaccine.

b) Phagocytic defect

- Supportive:
 - o Prophylactic antibiotics
 - o Prophylactic antifungal
 - o Gamma interferon in chronic granulomatous disease
- Definitive:

c) T cell defect

- Supportive:
 - o Aggressive management of established infections
 - o Antibiotic/ antifungal prophylaxis + IVIG therapy
 - o Irradiated , CMV neg blood if needed
 - o Withhold all live vaccines
 - o Infectious precautions when hospitalized
- Definitive:
 - o Bone Marrow Transplant: >80 -90% success iftransplanted early

d) B cell defect

- IVIG Replacement Therapy:
 - o Dose: 400-600mg/kg q4wk à aiming for troughlevels 5-6 g/L
 - o Side effects:allergic rxn, Transmission ofinfections, thromboembolic events, Renal complications
 - o Monitor:
 - Trough IgG levels pre each infusion (x6 infusion toreach steady trough level)
 - Q3-6 mo renal/ liver functions
 - Q1-3 yrs viral screen (PCR) – Hep B, C and HIV

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
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